

HYOUNGWON SEO

☎ +82 10-7200-2360 | ✉ guddnjs2366@gmail.com | Artificial Intelligence & Media Lab. (AIMLab)

Education

Hanbat National University (Advisor: Prof. Haneol Jang)

M.S. Student in Computer Engineering
GPA 4.5/4.5

Daejeon
Mar. 2025 - Present

Hanbat National University (Advisor: Prof. Haneol Jang)

B.S. in Computer Engineering
GPA 3.97/4.5 (Major: 4.12/4.5)

Daejeon
Mar. 2019 - Feb. 2025

Research Experience

Datamaker

Research Intern

Daejeon, Republic of Korea
Jul. 2024

- Conducted research on Korean document OCR and table recognition
- Developed a prototype for extracting text and tables from PDF documents

Publications

Conference

- Hyungwon Seo**, Dongsu Kim, Seoyeon Oh, Jisang Lee, and Haneol Jang. “Effective Detection of Generated Images Using Fast Fourier Transform and Discrete Cosine Transform”, *kdfs2023*, November 2023.
- Jisang Lee, **Hyungwon Seo**, and Haneol Jang. “Crop-Paste Technique for Enhancing Object Detection Performance in Synthetic Aperture Radar Images”, *kcc2024*, June 2024.
- Hyungwon Seo**, Jisang Lee, and Haneol Jang. “Data Preprocessing and Augmentation Methods for SAR to Optical Image Translation”, *kcc2024*, June 2024.

Journal

- Hyungwon Seo**, Dongsu Kim, Seoyeon Oh, Jisang Lee, and Haneol Jang. “Effective Detection of Generated Images Using Frequency Transform”, *Journal of Korean Institute of Information Scientists and Engineers*, vol. 52, no. 4, April 2025.
- Jisang Lee, **Hyungwon Seo**, Woojin Park, Haechul Choi, Kyuheon Kim, Eun-Kyung Kim, and Haneol Jang. “Crop-Paste Data Augmentation Technique for Enhancing Object Detection Performance in Synthetic Aperture Radar Images”, *Journal of Institute of Korean Electrical and Electronics Engineers*, vol. 29, no. 2, May 2025.
- Hyungwon Seo**, Jisang Lee, Woojin Park, Haechul Choi, Kyuheon Kim, and Haneol Jang. “Effective Data Preprocessing and Augmentation Methods of SAR Images for Optical Image Translation”, *Journal of Broadcast Engineering*, vol. 30, no. 3, May 2025.
- Suan Kim, **Hyungwon Seo**, Cheoneum Park, and Haneol Jang. “Multispectral Wildfire Detection via SWIR/NIR Band Optimization and Context-Aware Data Augmentation”, *Journal of Institute of Korean Electrical and Electronics Engineers*, vol. 29, no. 4, December 2025.

Research Projects

RAG-Based ChatAI for Cybersecurity Policy Analysis and Knowledge Utilization

National Security Research Institute

Jun. 2023 - Present

- Built automated pipelines for collecting and managing cybersecurity policy documents, and developed document classification and summarization models.
- Developed a RAG-based ChatAI system for policy question answering, source citation, and policy comparison.

SAR-to-Optical Image Translation Using Generative Models

Independent Research

Dec. 2024 - May 2025

- Studied diffusion- and GAN-based approaches for translating SAR imagery into optical images.
- Investigated preprocessing and geometric augmentation strategies for paired SAR and optical images.

Infrared Object Detection Using IR-to-RGB Translation

Independent Research

Jun. 2025 - Feb. 2026

- Developed an end-to-end framework combining one-step IR-to-RGB image translation with object detection.
- Investigated detector-oriented feature consistency using paired IR and RGB images while requiring only IR input at inference.

Infrared Object Detection Using Training-Time Visible Latent Guidance

Independent Research

Mar. 2026 - May 2026

- Learned visible-like latent representations from paired visible images without reconstructing full RGB images.
- Designed object-aware multi-scale feature injection for IR-only object detection.

SAR Building Segmentation Using Reliability-Guided EO Supervision

Independent Research

Jun. 2026 - Present

- Developed reliability-guided input construction and feature refinement for SAR building segmentation.
- Investigated training-time EO foundation-model supervision while preserving SAR-only inference.

Awards and Honors

Competition Award	4th Prize(SW 중심대학협의회 회장상), <i>SWuniv AI Competition 2023</i> (Satellite Image Building Area Segmentation)	Aug. 2023
Best Paper Award	President’ s Award(학회장상), Effective Detection of Generated Images Using Fast Fourier Transform and Discrete Cosine Transform, <i>Korean Digital Forensics Society 2023</i>	Dec. 2023
Competition Award	Grand Prize, <i>Research and Development Special Zone Promotion Foundation</i> (Global Wildfire Detection Challenge: Leveraging AI for Enhanced Satellite Imagery Analysis)	Apr. 2024
Competition Award	Gold Prize, <i>Capstone Design Exhibition, Department of Computer Engineering, Hanbat National University</i> (Team name: SkyPixel)	Nov. 2024

Technical Skills

Programming	Python, Java, C, C++
Professional Softwares	PyTorch, TensorFlow
Languages	Korean(Native), English(Beginner)

Teaching Experiences

Python Programming	Hanbat National University
Teaching Assistant (Prof. Haneol Jang)	Mar. 2025 - Jun. 2025
Computer Vision	Hanbat National University
Teaching Assistant (Prof. Haneol Jang)	Sep. 2025 - Dec. 2025
Understanding and Applications of Artificial Intelligence	Hanbat National University
Teaching Assistant (Prof. Haneol Jang)	Mar. 2026 - Jun. 2026